Channel Knowledge Maps for FR3 UAV Base Stations: Dataset and ML-Based Modeling

Guillem Moreno Garcia, Sergi Abadal, Evgenii Vinogradov
Universitat Politècnica de Catalunya (UPC), Barcelona, Spain
guillem.moreno.garcia@estudiantat.upc.edu

***Abstract*—This paper introduces a dense FR3 channel knowledge map (CKM) dataset for unmanned aerial vehicle base stations. It contains 16180 ray-traced layout–height combinations from 3236 urban locations in 66 cities at 7.125 GHz. Every 513×513 realization stores building geometry, geometric visibility, attenuation, delay spread, angular spread, and a variable transmitter height. As a focused first benchmark, we propose an interpretable height-aware attenuation model. Its line-of-sight branch combines free-space loss with a coherent image-source two-ray term and compact height/range calibration. Its non-line-of-sight branch combines a COST231-shaped basis with local morphology and topology-specific linear calibration. Under a strict city holdout, the frozen model reaches 1.9289 dB overall root mean square error, with 1.7370 dB in LoS and 3.5363 dB in NLoS, on 2590 maps from 14 unseen cities. Its median CPU time is 0.0602 s per map. The result provides a transparent attenuation baseline and a physical anchor for later distribution-aware refinement of all three channel quantities.**

***Index Terms*—channel attenuation, channel knowledge map, delay spread, path loss, radio map, unmanned aerial vehicle.**

I. Introduction

Uncrewed aerial vehicles (UAVs) carrying base stations can support future 6G networks through mobility and an increased probability of line of sight (LoS). Their deployment nevertheless requires fast, location-dependent channel information for placement and resource management. Channel knowledge maps (CKMs) address this need by indexing channel information by transmitter or receiver location [1], [2].

Dense CKMs can be measured or simulated. Measurements are costly to collect at scale, whereas deterministic ray tracing (RT) provides spatially complete labels at substantial computational cost. Learned surrogates such as RadioUNet and PMNet therefore use convolutional encoder–decoders to predict dense maps from city geometry and transmitter information [3], [4]. Existing datasets, however, predominantly use fixed terrestrial transmitters and omit spatially varying building height.

The upper mid-band frequency range, FR3, is being studied as a coverage/capacity bridge for 6G [5]. It is attractive for UxNB deployment, but blocked links remain relevant even with elevated transmitters. Assessing this setting requires height-aware channel data across diverse urban environments. To our knowledge, no public CKM dataset currently combines FR3, UxNB-specific geometry, and dense three-dimensional building information.

a) Limitations of the existing CKM datasets: Table I compares the learning inputs distributed by each dataset rather than the provenance of its source geometry. Here, 3236 location groups paired with five transmitter heights produce 16180 realizations. "Building-height raster" means that every building pixel retains a height, rather than only a binary footprint; it does not imply that every height was surveyed directly. We combine global products providing both footprint and height [9], [10], avoiding the sparse height tags in OpenStreetMap [11].

A UxNB CKM should cover a typical cell diameter of 300–500 m with a spatial resolution capable of capturing relevant variations over approximately 20–80 wavelengths [12]. Since UxNB heights can approach 500 m, their channels differ substantially from fixed terrestrial links. UrbanRadio3D includes receiver heights up to 20 m [6], but does not provide the continuously varying elevated-transmitter geometry considered here. Accurate building height is particularly important because it controls visibility and rooftop blockage.

b) Limitations of the existing CKM modeling approaches: Dense attenuation is commonly treated as unconstrained image-to-image regression. This asks large networks to relearn stable propagation structure, often under fixed terrestrial geometry, and obscures whether they transfer to unseen cities and continuously varying UxNB heights.

c) Beyond state-of-the-art contributions: In this paper, we address these gaps. The main contribution thus is two-fold:

1) We introduce a 7.125 GHz Channel Knowledge Map dataset generated via ray-tracing simulations in 3,236 city layouts across 66 diverse cities. This dataset relies on global-scale 3D building data [9], [10]. UxNB applications are enabled by using five transmitter heights per layout, resulting in 16180 CKM samples.
2) We propose a height-aware channel prediction framework[1] that separates LoS and NLoS propagation using local city geometry. Free-space, coherent two-ray, and COST231/Hata-derived terms are calibrated through compact lookup tables and ridge regression, enabling zero-shot transfer to unseen cities.

Notation: Bold uppercase symbols denote rasters or tensors, bold lowercase symbols denote vectors, and $\odot$ is element-wise multiplication.

II. CKM: Definition and Dataset Description

Figure 1 illustrates one sample from the dataset. In the following, UxNB and transmitter are used interchangeably.

[1]The dataset contains attenuation, angular spread, and delay spread; this first benchmark predicts attenuation only.

TABLE I
COMPARISON OF DENSE CKM AND RADIO-MAP DATASETS.

Property	This work	UrbanRadio3D [6]	RadioMapSeer [7]	CKMImageNet [8]
Variable elevated Tx height	yes	no	no	no
Layout groups or maps	3236 groups	701	701	40000
Raster size(s)	513^2	256^2	256^2	32^2–128^2
Horizontal resolution	1 m	1 m	1 m	2 m
Channel quantities	Att. (PL), LoS, DS, AS	PL, DoA, ToA	PL	Gain, AoA/AoD, delay
Building-height raster included	yes	yes	no	no
Per-sample topology labels	3 and 6 classes	no	no	no

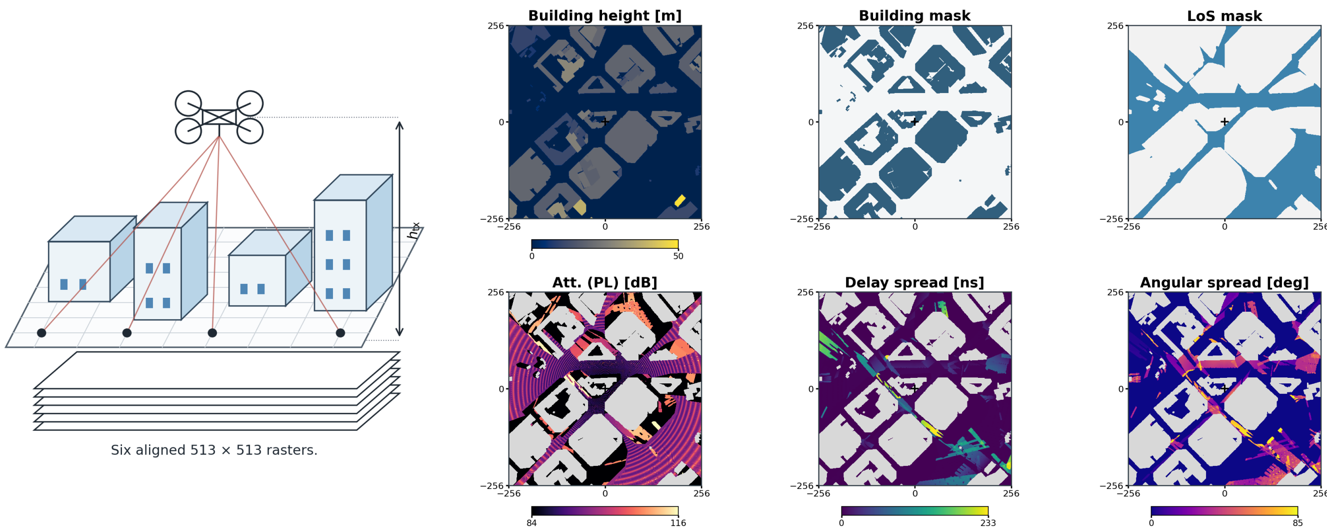

Fig. 1. One Barcelona realization (`sample_02806`) at $h_{\mathrm{tx}} = 69.4$ m. The left illustration shows an elevated UxNB above six aligned 513×513 rasters. The panels show building height, building mask, LoS, attenuation, delay spread, and angular spread. Crosses mark the centered transmitter; gray areas are building pixels excluded from the metrics. For this sample, the stored three- and six-class labels are `dense_highrise` and `dense_block_midrise`.

A. CKM Definition and Mapping Formulation

For realization n, let $\mathbf{E}_n \in \mathbb{R}^{H\times W\times L}$ contain environmental layers, $\mathbf{C}_n \in \mathbb{R}^{H\times W\times F}$ contain channel quantities, and $\mathbf{m}_n \in \mathbb{R}^K$ contain metadata. CKM estimation seeks a mapping

$$\widehat{\mathbf{C}}_n = \mathbf{\Phi}_{\boldsymbol{\theta}}(\mathbf{E}_n, \mathbf{m}_n), \tag{1}$$

where $H = W = 513$, $L = 2$, and $F = 4$.

B. The FR3 UxNB CKM Dataset

We construct a comprehensive benchmark dataset using high-precision deterministic simulations.

1) 3D urban environment modeling ($\mathbf{E}$)*:* The environmental tensor $\mathbf{E}_n$ combines GlobalBuildingAtlas and 3D-GloBFP [9], [10]. Unlike optional OpenStreetMap height tags, whose availability is below 10% [11], these products provide both building footprint and height. This information is necessary to represent how UxNB altitude changes rooftop blockage and visibility.

We extract city layouts of 3,236 urban locations in 66 diverse cities worldwide to cover a range of propagation topologies (from open low-rise suburbs to dense high-rise downtown areas). Each rasterized environmental map $\mathbf{E}_n$ spans a spatial grid of $H = W = 513$ at a horizontal resolution of 1 m. The $L = 2$ environmental channels encode:

1) Building height profile maps $\mathbf{H}$ indicating building height in meters (0 m over ground level).
2) Binary building occupancy masks $\mathbf{B}$ indicating presence of structural obstacles.

2) Ray-Tracing channel simulation ($\mathbf{C}$)*:* High-precision ray-tracing simulations are conducted to generate the spatially consistent channel tensors $\mathbf{C}_n$. To avoid using separate software for 3D model construction (e.g., Blender) and for RT (e.g., Wireless InSite or Sionna), SiteViewer and RT capabilities of MATLAB are used in this work.

RT simulation parameters are summarized in Table II. The 7.125 GHz frequency is a WRC-23 candidate band available across all ITU regions, while the 1 m resolution corresponds to approximately 24 wavelengths [12]. The UxNB is horizontally centered and evaluated at five independently sampled heights per location. The observed range is approximately 12 m to 478 m, yielding $N = 16180$ samples. Receivers are placed at 1.5 m over every non-building coordinate. The channel tensor stores $F = 4$ channels:

1) Binary visibility mask $\mathbf{V}$: LoS (1) vs. NLoS (0) states.
2) Channel attenuation in dB.
3) Multi-path RMS delay spread in ns.
4) Multi-path RMS angular spread in degrees.

Building elements are assigned zero in $\mathbf{C}_n$ and are excluded from training and evaluation.

3) Metadata ($\mathbf{m}$)*:* Because carrier frequency, receiver height, and horizontal transmitter location (fixed at the grid

TABLE II
RAY-TRACING SIMULATION PARAMETERS.

Parameter	Value
Carrier frequency, f	7.125 GHz
Antenna Model	Isotropic
Sample dimensions	513×513 at 1 m
Ground receiver height, h_{rx}	1.5 m
Observed transmitter height, h_{tx}	12 m to 478 m
UxNB Heights per Location	5
Total CKM Samples, N	16,180

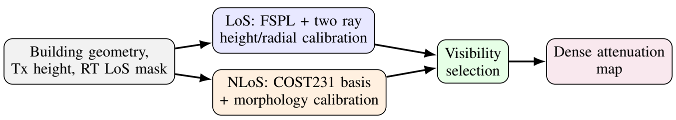

Fig. 2. Separate LoS and NLoS mechanisms are calibrated on training cities.

center) are constant across all simulations, the metadata vector $\mathbf{m}_n$ consists of:

1) Transmitter height h_{tx}, observed from approximately 12 m to 478 m.
2) Categorical tags mapping the layout to three- and six-class morphology partitions. The attenuation model re-computes its three-class ITU route from the raster as described in Section III-B.

The labeled HDF5 file stores both three- and six-class topology tags and the transmitter-height bin alongside the six rasters. Generation and model code are publicly available.[2]

III. HEIGHT-AWARE CKM MODELING

Figure 2 summarizes the proposed attenuation-focused mapping. Only compact lookup tables and linear weights are learned, and all are frozen before a held-out city is evaluated. On valid ground receivers, the stored visibility mask $\mathbf{V}$ selects

$$\boldsymbol{\Lambda} = \mathbf{V} \odot \boldsymbol{\Lambda}_{\text{LoS}} + (1 - \mathbf{V}) \odot \boldsymbol{\Lambda}_{\text{NLoS}}, \tag{2}$$

where $\odot$ denotes element-wise multiplication. The mask is the configured MATLAB ray tracer's binary visibility output, not a prediction of the attenuation model. For matching geometry and RT settings, it can be stored once and reused.

Given the deployment geometry (h_{tx}, h_{rx}, and the fixed horizontal UxNB position), we pre-compute spatial geometry matrices over the 513×513 grid: horizontal distance $\mathbf{D}_{\text{2D}}$, direct 3D distance $\mathbf{D}_{\text{3D}}$, ground-reflected path length $\mathbf{D}_{\text{GR}}$, and elevation angle $\boldsymbol{\Theta}_{\text{2D}}$. For any receiver location $p = (i, j)$, the scalar geometric parameters $(d_{\text{2D}}, d_{\text{3D}}, d_{\text{GR}}, \theta)$ are indexed directly from these matrices. For notational brevity, we omit the spatial index p in subsequent expressions.

A. Line-of-Sight Branch

For LoS propagation, the attenuation model combines free-space path loss Λ_{FSPL} with a coherent ground-reflection term C_{2ray} and CKM-calibrated height and radial corrections:

$$\Lambda_{\text{LoS}} = \Lambda_{\text{FSPL}}(d_{\text{3D}}) + C_{\text{2ray}} + \zeta(h_{\text{tx}}) + r(d_{\text{2D}}, h_{\text{tx}}), \tag{3}$$

[2] https://github.com/unworthyzeus/CKMGenerator; https://github.com/unworthyzeus/Final_Code_TFG.

where $\zeta(h_{\text{tx}})$ is a height-dependent scalar bias correction and $r(d_{\text{2D}}, h_{\text{tx}})$ is a residual component capturing the effects introduced by horizontal distance and transmitter height.

The free-space term follows the Friis distance dependence, while the coherent correction retains the phase difference between the direct and reflected paths [13]:

$$\Lambda_{\text{FSPL}} = 20 \log_{10}\left(\frac{4\pi d_{\text{3D}} f}{c}\right), \tag{4}$$

$$C_{\text{2ray}} = -20 \log_{10}\left|1 + \rho(h_{\text{tx}}) \frac{d_{\text{3D}}}{d_{\text{GR}}} e^{-j[k(d_{\text{GR}} - d_{\text{3D}}) + \phi(h_{\text{tx}})]}\right|, \tag{5}$$

where k is the free-space wavenumber, ρ is an effective reflected-field amplitude, and ϕ is a fitted phase offset. The calibration terms are obtained in two steps. Firstly, in each 5 m height bin, candidates $\hat{\rho} \in \{0, 0.05, \ldots, 1.5\}$ and 48 uniformly spaced phases $\hat{\phi}$ in $[-\pi, \pi)$ are evaluated. For a candidate pair, the remaining bias is calculated as

$$\hat{\zeta}(\hat{\rho}, \hat{\phi}) = \frac{1}{|B_{\text{LoS}}|} \sum_{i \in B_{\text{LoS}}} \left[y_i - \Lambda_{\text{FSPL},i} - C_{\text{2ray},i}(\hat{\rho}, \hat{\phi})\right], \tag{6}$$

where B_{LoS} is the set of valid LoS receivers and y is the target attenuation. The combination of $\hat{\zeta}, \hat{\rho}, \hat{\phi}$ minimizing the bin root mean square error (RMSE) is retained. Secondly, the remaining radial calibration term $r(d_{\text{2D}}, h_{\text{tx}})$ is calculated as the mean of the residual error.

In implementation terms, the LoS calibration is a lookup table that contains ρ, ϕ, ζ and r. Inference linearly interpolates neighboring height and radial entries.

B. Non-Line-of-Sight Branch

For shadowed receivers, an empirical COST231/Hata-derived basis is augmented by local structural features from $\mathbf{E}$. The spatial context vector $\mathbf{x} \in \mathbb{R}^{14}$ is evaluated with a topology- and height-dependent calibrated weight vector $\hat{\mathbf{w}}_q$.

Evaluated at $f_{\text{MHz}} = 7125$, the COST231/Hata-derived expression is used only as an inspectable basis because its source domain does not cover the present frequency or heights [14]. With $d_{\text{km}} = \max(d_{2D}/1000\,\text{m}, 0.001)$,

$$\begin{aligned}\widetilde{\Lambda}_C = {} & 46.3 + 33.9 \log_{10} f_{\text{MHz}} - 13.82 \log_{10} h_{\text{tx}} - a(h_{\text{rx}}) \\ & + (44.9 - 6.55 \log_{10} h_{\text{tx}}) \log_{10} d_{\text{km}} + 3, \end{aligned} \tag{7}$$

$$a(h_{\text{rx}}) = (1.1 \log_{10} f_{\text{MHz}} - 0.7) h_{\text{rx}} - (1.56 \log_{10} f_{\text{MHz}} - 0.8). \tag{8}$$

The regression feature is $\Lambda_C = \text{clip}(\widetilde{\Lambda}_C, 0, 185)$.

To account for the local environment, we derive 13 contextual features from the building height grid $\mathbf{H}$ and building mask $\mathbf{B}$. Let $\mathbf{G} = 1 - \mathbf{B}$ mark ground and $\mathbf{N} = (1 - \mathbf{V}) \odot \mathbf{G}$ mark NLoS ground. For a reflect-padded $k \times k$ box mean $\mathcal{A}_k$, $k \in \{15, 41\}$, define

The full feature vector $\mathbf{x}$ is constructed as:

$$\begin{aligned}\mathbf{x} = [& \Lambda_C, \ell_d, \delta_{15}, \delta_{41}, h_{15}, h_{41}, \delta_{41}\ell_d, n_{15}, n_{41}, \\ & n_{41}\ell_d, \sigma_{\text{sh}}, \theta_n, n_{41}\theta_n, 1]^\top. \end{aligned} \tag{9}$$

where

- $\ell_d = \ln(1 + d_{2D}/1\,\mathrm{m})$ is log-radial growth;
- $\delta_k = \mathcal{A}_k[\mathbf{B}]$, $h_k = \mathcal{A}_k[\mathbf{H} \odot \mathbf{B}]/90\,\mathrm{m}$, and $n_k = \mathcal{A}_k[\mathbf{N}]$ describe local and regional morphology;
- $\theta_n = \mathrm{clip}(\theta/90, 0, 1)$ and $\sigma_{\mathrm{sh}} = 2.3197[\max(90 - \theta, 0)]^{0.2361}$ describe elevation; and
- the remaining products are density–range, blockage–range, and blockage–elevation interactions, followed by the intercept.

Routing follows the standard ITU topology and transmitter-height partitions [15], [16]. Let $\mathcal{B}$ be the building pixels, let their four-connected components represent N_b footprints, let $A = 513^2 \times 10^{-6}$ km^2, and let H_j be the maximum raster height in footprint j. We estimate

$$\hat{\alpha} = \frac{|\mathcal{B}|}{513^2}, \quad \hat{\beta} = \frac{N_b}{A}, \quad \hat{\gamma} = \sqrt{\frac{1}{2N_b} \sum_{j=1}^{N_b} H_j^2}. \tag{10}$$

Thus, $\hat{\alpha}$ is the built-up fraction, $\hat{\beta}$ is the footprint density in buildings per km^2, and $\hat{\gamma}$ is the fitted Rayleigh mode of the footprint heights. For prototype k, the selector computes $d_k = \|\log[(\hat{\alpha}, \hat{\beta}, \hat{\gamma})/(\alpha_k, \beta_k, \gamma_k)]\|_2$ and chooses the smallest. Equal-weight log-ratios compare relative deviations and prevent the scale of β from dominating. The ITU prototypes are suburban $(0.1, 750, 8)$, urban $(0.3, 500, 15)$, dense urban $(0.5, 300, 20)$, and urban high rise $(0.5, 300, 50)$. Urban high rise is merged into dense urban because only 90 training maps select it. Independently, transmitter height is low for $h_{\mathrm{tx}} \le 60\,\mathrm{m}$, middle for $60\,\mathrm{m} < h_{\mathrm{tx}} \le 100\,\mathrm{m}$, and high above $100\,\mathrm{m}$. The two labels select one of nine fitted coefficient sets without using the city name or channel target.

For each training map, at most 1024 NLoS pixels are selected deterministically. Non-intercept features are standardized within regime q and fitted by ridge regression:

$$\hat{\mathbf{w}}_q = \arg\min_{\mathbf{w}} \sum_{i \in q} (\mathbf{z}_i^\top \mathbf{w} - y_i)^2 + 10^{-2} \|\mathbf{w}_{\neg b}\|_2^2. \tag{11}$$

The intercept is unpenalized. After transforming the coefficients back to raw feature space, the deployed estimate is

$$\Lambda_{\mathrm{NLoS}} = \mathrm{clip}(\mathbf{x}^\top \hat{\mathbf{w}}_q, 20, 185). \tag{12}$$

IV. Results

A. Protocol

The city holdout contains 10840 maps from 37 training cities, 2750 from 15 validation cities, and 2590 from 14 final-test cities. Calibration uses training cities only; validation is reserved for development and ablation. Building pixels are excluded before receiver statistics or metrics are computed. The final test metrics contain 423 087 119 valid non-building receivers: 391 721 805 LoS and 31 365 314 NLoS. NLoS accounts for 7.413% of evaluated pixels, although 424 test maps contain more than 25% valid NLoS receivers.

For valid mask $m_i(p)$, estimate $\hat{y}_i(p)$, and target $y_i(p)$,

$$\mathrm{RMSE} = \sqrt{\frac{\sum_i \sum_p m_i(p)[\hat{y}_i(p) - y_i(p)]^2}{\sum_i \sum_p m_i(p)}}. \tag{13}$$

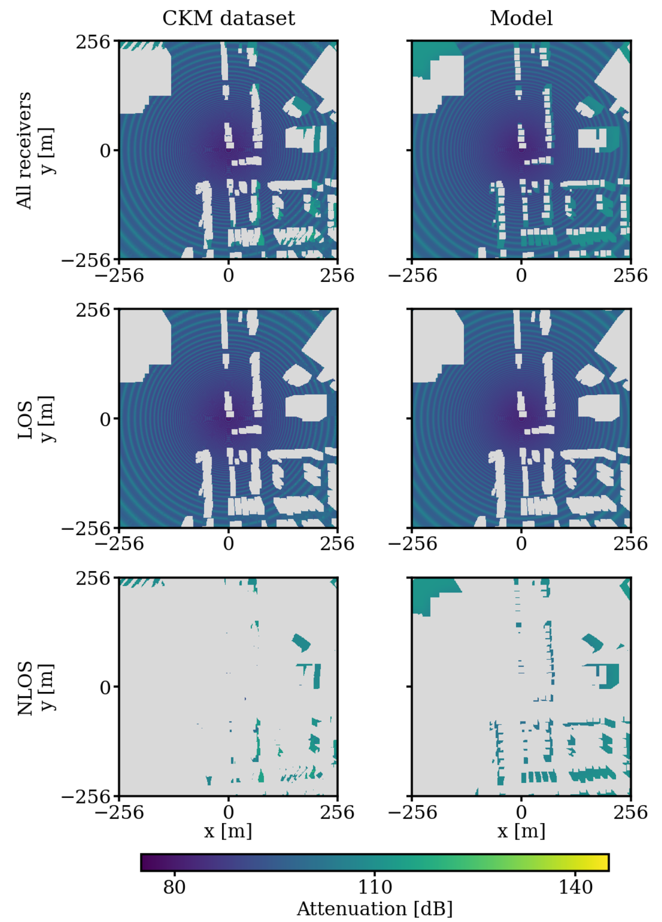

Fig. 3. Held-out Vancouver `sample_15142` ($h_{\mathrm{tx}} = 52.22\,\mathrm{m}$; suburban, low-height regime). The dataset column shows only stored valid targets, while the model column predicts every ground receiver. Thus, gray in the first dataset panel denotes buildings or an unavailable target; in the first model panel it denotes buildings only. Lower rows additionally exclude the other visibility class. Metrics use valid targets only. The map has 198083 LoS and 12614 NLoS targets; full, LoS, and NLoS RMSE are 1.85, 1.60, and 4.11 dB.

TABLE III
FROZEN TEST METRICS; INTERVALS RESAMPLE 14 CITIES.

Metric	Estimate	95% interval
Overall RMSE [dB]	1.9289	[1.7985, 2.0960]
Overall MAE [dB]	1.2170	[1.1371, 1.3177]
LoS RMSE [dB]	1.7370	[1.6603, 1.8225]
NLoS RMSE [dB]	3.5363	[3.2335, 3.8482]
Macro-city RMSE [dB]	1.9831	[1.8495, 2.1297]

B. Ablation and Accuracy

Validation ablation confirms that the coherent reflected field is essential: adding it to FSPL with height bias reduces LoS RMSE from 3.7900 to 1.7412 dB. In NLoS, training-only global calibration reduces the raw COST231-basis RMSE from 25.4045 to 3.2179 dB; observable topology/height routing reaches 3.1983 dB. Thus, calibration explains the main NLoS reduction, while routing provides a smaller refinement.

On final test, the complete model reaches 1.9289 dB overall

TABLE IV
PIXEL-WEIGHTED TEST RMSE [dB] BY FITTED ITU REGIME.

Regime	Maps	Overall	LoS	NLoS
Low height	899	2.1437	1.7923	3.7526
Middle height	773	1.9682	1.7747	3.5080
High height	918	1.7117	1.6697	2.7620
Suburban	1455	1.6977	1.6048	3.1632
Urban	940	2.2963	1.9967	3.6466
Dense urban	195	2.5716	2.1126	4.1319
All	2590	1.9289	1.7370	3.5363

TABLE V
REPORTED RMSE [dB]. ONLY THE FIRST TWO ROWS SHARE A TEST CONTRACT.

Method	Evaluation contract	All	LoS	NLoS
Proposed	CKM city holdout	1.9289	1.7370	3.5363
GMM U-Net	Same split and masks	4.3606	4.4323	3.3381
RadioUNet [3]	RadioMapSeer DPM	1.60	–	–
RadioVAE$_C$ [17]	RadioMapSeer DPM	1.42	–	–
TSRDiff [18]	BRAT-Lab, 10% samples	3.62	–	–
RO-HMTGP [19]	Measured RSRP	4.19–6.00	–	–

RMSE and 1.2170 dB MAE; LoS and NLoS RMSE are 1.7370 and 3.5363 dB. A 20000-draw city bootstrap gives the interval in Table III. Per-city RMSE ranges from 1.6149 dB in Halifax to 2.5851 dB in Osaka. A compact model retaining eight of the 14 features preserves over 90% of the complete model's improvement over a single global NLoS constant.

Table IV uses the standard ITU low, middle, and high height bins fitted by the model. Error decreases with height, particularly in NLoS. The topology rows show materially different regimes: dense urban NLoS is hardest, while suburban maps are most accurate. The fitted coefficient vectors also change substantially across topologies, so one global set of slopes would mix distinct relationships.

The GMM U-Net is a controlled neural alternative: a height-conditioned mixture head and U-Net decoder, separated into six topology classes and LoS/NLoS experts, are evaluated on the same 2590 maps and masks. It was the best purely neural approach in an internal exploration of over 50 configurations. Its lower NLoS error shows where distribution-aware learning can help, while the lower LoS and overall errors of the proposed model show the value of resolving the stable coherent component explicitly. A following paper will study a hybrid model using calibrated attenuation, delay spread, and angular spread priors as inputs and base estimates for neural refinement.

The remaining rows are published context, not a controlled ranking. RadioUNet and RadioVAE use 256^2 fixed-height terrestrial RadioMapSeer data; TSRDiff uses sparse targets, and RO-HMTGP predicts measured RSRP rather than dense simulated attenuation. Their values indicate scale under different tasks, not superiority over the present 513^2 variable-height city holdout.

TABLE VI
COMPUTATIONAL CONTEXT; REPORTED CPU ROWS USE THE SAME CPU.

Method	Grid	State	Reported MAC	CPU [s]
Proposed	513^2	66.8k+126	0.170M*	0.0602
RadioUNet [3]	256^2	13.27M	12.73G	0.1339
PMNet [4]	256^2	33.34M	52.71G	0.4042
SSL-Radio [20]	256^2	15.4–20.4M	16.9–21.9G†	–
MATLAB RT	513^2	–	–	102.289

C. Computational Comparison

Table VI uses the same AMD Ryzen 5 5600X, six CPU threads, batch one, and GPU-disabled execution for the proposed model and neural baselines. The neural timings use official implementations, 10 warmups, and 50 forward passes; input and file I/O are excluded.

The asterisk counts the final 14-term NLoS dot product averaged over valid test NLoS pixels; feature construction and lookup operations are additional. The dagger is reconstructed from SSL-Radio's layer table under two plausible skip concatenations rather than measured from released code. The proposed timing includes feature construction and LoS evaluation. MATLAB R2024b takes a median 102.289 s over five Barcelona layouts on the same CPU. RT also produces visibility and all three channel quantities, so the raw $\sim 1700\times$ difference is not a like-for-like speedup. It nevertheless demonstrates a significant computational gap for the attenuation model.

D. Scope and Limitations

The reported accuracy is conditional on the supplied RT visibility mask. For fixed city geometry, transmitter position and height, receiver grid, and RT settings, that mask is deterministic and can be stored once; a repeated five-condition audit found all 742392 receiver visibility values identical. The fitted tables also inherit the simulator, isotropic antenna, fixed carrier, flat terrain, and building products used to create the data. COST231 supplies only a calibrated basis outside its original domain, and the model does not identify the particular reflected or diffracted NLoS path. These constraints are why the result is presented as a transparent dataset baseline and refinement anchor, not as a universal propagation law.

V. CONCLUSIONS

We introduced a dense FR3 UAV CKM dataset containing attenuation, delay spread, angular spread, visibility, building geometry, transmitter height, and topology labels for 16180 realizations from 66 cities. The accompanying attenuation benchmark separates coherent LoS structure from calibrated NLoS morphology and freezes every fitted quantity before unseen-city evaluation. It reaches 1.9289 dB overall RMSE while remaining compact, auditable, and fast on a CPU. The results further show that topology is the principal NLoS calibration partition, with transmitter height refining that partition and the regional morphology scale carrying most of the useful local context.

The model is intended as a physical/statistical base rather than a rejection of deep learning. The controlled GMM U-Net result identifies NLoS as the promising refinement target, but also shows the cost of asking a network to relearn the coherent LoS field. A future journal paper will train distribution-aware networks to refine analogous calibrated priors for attenuation, delay spread, and angular spread. Its multi-output runtime will not be directly comparable to this attenuation-only benchmark, but the present computational gap shows that the prior-first strategy is significant and worth preserving.

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